

Course Assignment

Lecture 1: Assignments

Tamás Grósz



The Assignment

The goal is to develop an AI system **end-to-end**.

Raw Data → Data Pipeline → ML Pipeline → Model → Scalable Inference

The project is divided into four milestones:

- ① Data source, task, splits, and storage
- ② Data analysis and data pipeline
- ③ ML pipeline, MLOps, and first models
- ④ Scalable inference and continued deployment

From Data to Production

M1	M2	M3	M4
Data Source	Data Pipeline	ML Pipeline	Serving & Deployment

- Each milestone builds on the previous one.
- The final system should demonstrate an integrated workflow.
- The emphasis is not only on model accuracy.
- **Data handling, reproducibility, automation, scalability, and deployment matter.**

Milestone 1 — Data Source and Task

Goal: Establish a solid foundation for the project.

- ① Select a suitable **raw data source**.
- ② The dataset should contain:
 - At least **10,000 learning samples**.
 - Realistic imperfections.
 - Missing values and/or other data quality issues.
- ③ Define a meaningful **AI/ML task**.
- ④ Identify the target variable and relevant features.

Think beyond “find a dataset”

Choose data that gives you enough room to demonstrate data handling, infrastructure, and ML practices.

Milestone 1 — Where to Find Data?

There are many sources for suitable datasets. Some places to start:

- **Kaggle**: datasets and competitions across many application areas
- **Hugging Face Datasets**: text, image, audio, multimodal, and other AI datasets
- **data.gov.ie**: Irish public-sector and open government data
- **Common Crawl**: large-scale web crawl data for text and web-based AI applications

Other possibilities

- Open data portals from governments and public organisations
- Research datasets
- Data generated or collected as part of your own project

Remember

Choose public data and always check the EULA (avoid legal issues)

Milestone 1 — Evaluation Strategy

Define how the model will be evaluated.

Choose and justify:

- Train / development / test split
- Cross-validation

Important

The evaluation strategy should match the nature of your task and the way the system will eventually be used.

Milestone 1 — Storage Solution

Decide how the project's data will be stored.

Consider:

- Where will the raw data live?
- Where will processed data be stored?
- What file formats will you use?
- Do you need a database or object storage?
- How will different versions of the data be identified?
- How will the system access the data?

Deliverable

A justified data source, ML task, evaluation strategy, and initial storage architecture.

Milestone 2 — Data Analysis

Goal: Understand the data before building the model.

Perform an exploratory analysis covering topics such as:

- Data distributions
- Feature relationships
- Missing values
- Duplicates
- Outliers
- Invalid or inconsistent values
- Class imbalance
- Potential data leakage

Question to answer

What problems would a model encounter if we trained on the raw data?

Milestone 2 — Build the Data Pipeline

Turn the analysis into a repeatable data workflow.

Raw Data → Extract → Transform → Load

The pipeline should address issues such as:

- Missing-value handling
- Outlier handling
- Data type conversion
- Cleaning and validation
- Feature transformations
- Dataset generation

ELT / ETL: Choose an appropriate approach and justify it.

Milestone 3 — ML Pipeline and MLOps

Goal: Move from data preparation to reproducible model development.

Set up an ML pipeline containing appropriate stages, for example:

Data → Features → Training → Evaluation → Model

Introduce appropriate MLOps practices:

- Version control
- Reproducible experiments
- Configuration management
- Model/version tracking
- Automated training or evaluation where appropriate
- Logging and monitoring of experiments

Milestone 3 — First Model

Train and evaluate the **first version of the model**.

You should be able to answer:

- What model did you choose?
- Why is it appropriate for the task?
- What baseline are you comparing against?
- Which metrics are you using?
- How good is the first version?
- Can someone else reproduce the experiment?

The goal

This milestone is about establishing a working, reproducible ML development process, not necessarily producing the best possible model yet.

Milestone 4 — Scalable Inference

Turn the model into a usable service.

Client → Inference API → Model → Prediction

Consider:

- API design
- Containerization
- Deployment
- Scaling
- Resource requirements
- Latency and throughput
- Reliability

Key question

What happens when 1 user becomes 1,000 or 100,000 users?

Milestone 4 — Model Updates and Deployment

A production ML system does not stop after the first deployment.

Demonstrate a model update:

New Data → Retraining → New Model → Deployment

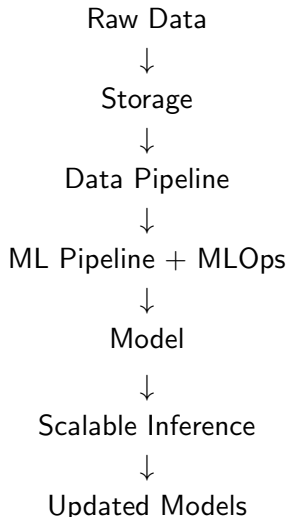
The project should demonstrate:

- Updating the model
- Versioning the new model
- Deploying the updated version
- Showing that the service now uses the updated model

Think of the project as a living system

Data changes → models change → infrastructure must adapt.

The Complete Project



Milestone Summary

Milestone	Main focus	Weight
M1	Data source, task, splits, storage	10%
M2	Data analysis and data pipeline	10%
M3	ML pipeline, MLOps, first model	10%
M4	Scalable inference and model updates	60%

Final project

The final submission should bring all four stages together into one reproducible AI system.

Don't just train a model.

Build the system around it.

Data → **Infrastructure** → **ML** → **Production**